KARTHEEK MUKKAVILLI

Austin, TX | 832-752-3951 | kartheek.mukkavilli@gmail.com | linkedin.com/in/kartheek-mukkavilli | github.com/mkskart | U.S. Citizen

EDUCATION

University of Texas at Austin Austin, TX

B.S. Electrical & Computer Engineering | *GPA: 3.95/4.0* | *Expected May 2028*

Coursework: Embedded Systems · Digital Logic Design · Digital Signal Processing · Discrete Mathematics · Software Design · Circuit Theory

Elements of Computing Certificate (CS) | Business Foundations Minor

TECHNICAL SKILLS

Languages: C · C++ · Python · Assembly

Embedded/RTOS: Zephyr RTOS · FreeRTOS · nRF Connect SDK · LTE-M · GNSS · LoRa · MQTT · UART · RTT · I2C · SPI · PWM · ESP-NOW · TLS/mTLS

Hardware/Tools: nRF9160 · ESP32 · Raspberry Pi · Arduino · MPU-6050 · KiCad · Fusion 360 · Saleae Logic Analyzer · nrfjprog · AWS IoT Core

Certifications: PCAP (Python Associate) · PCEP (Python Entry)

EXPERIENCE

National Oilwell Varco Houston, TX

Embedded Systems Intern *Jun – Aug 2025*

- Replaced $20K/site legacy PLC/Modbus installations with a custom nRF9160 board under $250 (98% cost reduction), deployed across 50 field units within a $5K R&D budget.
- Engineered C/Zephyr RTOS firmware for LTE-M + GNSS over a 40-node LoRa mesh to AWS IoT Core; wrote a safe-shutdown handshake preventing LTE tower rejection on power cycle.
- Automated TLS certificate provisioning via Python + AT%CMNG, reducing per-unit onboarding from hours to under one minute across all deployed units.
- Led full product lifecycle — KiCad PCB layout, Fusion 360 enclosure, firmware dev, and field validation — delivering production-ready hardware under budget.

Human Centered Robotics Laboratory — UT Austin Austin, TX

Research Assistant *Jan 2026 – Present*

- Architected C++ firmware on Raspberry Pi Pico to synchronize 16-DOF bionic hand kinematics from remote glove telemetry, achieving 1ms sensor-to-actuation latency.
- Tuned PID control loops for 10 independent joints, improving tracking from unreliable to consistently stable real-time replication across the full range of hand motion.

J.P. Morgan Chase Houston, TX

Software Engineering Intern *Jun – Jul 2022*

- Built a Spring Boot/Cassandra microservice for a 2M+ record keyspace; optimized SQL logic reducing latency by 40%.

PROJECTS

LiteWing — Custom Quadcopter Flight Controller | C++ · FreeRTOS · ESP32 · MPU-6050

- Programmed a 1 kHz FreeRTOS PID attitude control loop; measured 1.0ms loop period with 45µs jitter via GPIO toggling and Saleae logic analyzer.
- Complementary filter IMU fusion via I2C + 400Hz PWM motor control; achieved ±2.5° attitude hold over a 60-second flight test.
- Designed a custom ESP-NOW telemetry and command layer between flight controller and transmitter.

nrf9160-zephyr-template | C · Zephyr RTOS · nRF9160 · AWS IoT · LoRa

- Open-sourced a production Zephyr firmware template for the nRF9160: LTE-M bring-up with exponential backoff, GNSS PVT parsing, AWS IoT MQTT over mTLS, RTT/UART logging, and automated TLS provisioning — generalized from NOV production firmware.

pid-motor-sim — PID Control Simulator | Python

- 3-DOF DC motor arm simulation with anti-windup PID, derivative-on-measurement, and Ziegler-Nichols auto-tuning across an 80-config gain sweep with rise time, settling time, and overshoot metrics.

kinematic-transformer | Python · PyTorch

- Transformer encoder for real-time multi-joint kinematic forecasting; 0.571ms inference latency — demonstrates ML-to-actuation pipeline design for embedded control systems.

LEADERSHIP

UT IEEE RAS Engineer | TSA Co-Founder: 3× National Finals | 4× Gold PVSA Medalist | SAMPADA Violin Certified · ABRSM Grade 5